Probability Theory

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1 Probability Theory

Learning outcomes

- understand the concept of random variables
- understand the concept of probability
- understand and learn to use resampling to compute probabilities
- understand the concept probability mass function
- understand the concept probability density function
- understand the concept cumulative distribution functions
- use normal distribution
- understand the central limit theorem
- to understand and define sampling distribution and standard error
- to compute standard error of mean and proportions

2 Introduction to probability

Some things are more likely to occur than others. Compare:

- the chance of the sun rising tomorrow with the chance that no one is infected with Ebola next year
- the chance of a dark winter in Stockholm with the chance of no rainy days during the summer in Stockholm

We intuitively believe that the sun rising and a dark winter in Stockholm are much more likely than Ebola disappearing overnight or having no rain during the entire summer. **Probability** gives us a way to quantify how likely events are. **Probability rules** enable us to reason about uncertainty. The probability rules are expressed in terms of sets, i.e., a well-defined collection of distinct objects.

Suppose we perform a random experiment that we do not know the outcome of, i.e., we are uncertain about the outcome. We can, however, describe all possible outcomes of the experiment.

- The **sample space** is the set S of all possible outcomes. For example, when rolling a 6-sided die, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$.
- An **event** is a subset of the sample space.
- An event is said to **occur** if the outcome of the experiment belongs to that set.
- The **complement** of an event E, denoted E', contains all outcomes in S that are not in E.
- Two sets E and F, such that $E \cap F = \emptyset$, are said to be **disjoint**.

2.1 Axioms of probability

1. $0 \leq P(E) \leq 1$ for any event $E \subseteq S$
2. $P(S) = 1$
3. If E and F are disjoint events, then $P(E \cup F) = P(E) + P(F)$

2.2 Common rules of probability

Based on the axioms, the following rules of probability can be proved.

- **Complement rule**: let $E \subseteq S$ be any event, then $P(E') = 1 - P(E)$.
- **Impossible event**: $P(\emptyset) = 0$.
- **Probability of a subset**: let $E, F \subseteq S$ be events such that $E \subseteq F$, then $P(F) \geq P(E)$.
- **Addition rule**: let $E, F \subseteq S$ be any two events, then $P(E \cup F) = P(E) + P(F) - P(E \cap F)$.

2.3 Conditional probability

Let $E, F \subseteq S$ be two events with $P(E) > 0$, then the conditional probability of F given that E occurs is defined as

$$P(F \mid E) = \frac{P(E \cap F)}{P(E)}$$

.

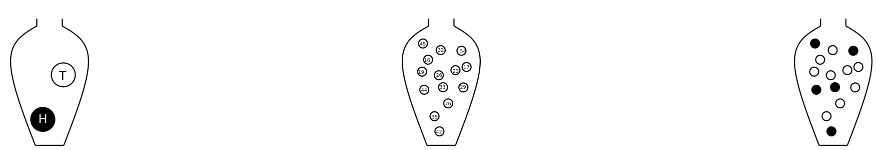
The **product rule** follows from conditional probability:

$$P(E \cap F) = P(F \mid E)P(E) = P(E \mid F)P(F)$$

2.4 The urn model

The urn model is a simple mathematical model commonly used in statistics and probability. In the urn model, objects (such as people, mice, cells, genes, or molecules) are represented by balls with different colors or labels.

For example, a fair coin can be represented by an urn with two balls, labeled H and T. A group of people can also be represented by an urn: if age is the variable of interest, we write each person's age on a ball; if we are instead interested in if the people are allergic to pollen or not, we color the balls according to allergy status.



(a) Fair coin (b) Age (c) Pollen allergy status

Figure 2.1: Urn models of a fair coin, age of a group of people, and pollen allergy status of a group of people.

In the urn model, every unit (ball) is equally likely to be selected. This means that the urn model is well suited to represent flipping a fair coin. However, a biased coin can also be modeled using an urn model by changing the number of balls that represent each side of the coin.

By drawing balls from the urn with or without replacement, probabilities and other properties of the model can be inferred. Drawing with replacement means that the draws are independent and the composition of the urn stays identical after each draw, whereas drawing without replacement means that the draws are dependent and the composition of the urn changes after each draw. For example, with an urn representing a population of people who are or are not allergic, we can calculate the probability of randomly selecting three people who are all allergic.

2.5 Random variables

A **random variable** is a variable whose value is determined by a random experiment.

A random variable cannot be predicted exactly before the experiment is performed, but its possible values and their probabilities can be described.

Random variables are usually denoted by capital letters, such as X, Y, and Z. Observed values of random variables are usually denoted by lowercase letters, such as x, y, and z.

The **population** is the full set of individuals or objects of interest, and a **sample** is a subset of the population.

Examples of random variables:

- the result of rolling a fair 6-sided die, D
- the weight of a random newborn baby, W
- the smoking status of a random mother, S
- the hemoglobin concentration of a random individual, Hb
- the number of mutations in a gene, M
- the BMI of a random man, B
- the weight status of a random man (underweight, normal weight, overweight, obese)

Examples of probabilities involving these random variables include:

- $P(W > 4.0 \text{ kg})$
- $P(S = 1)$
- $P(Hb < 125 \text{ g/L})$

Conditional probability can, for example, be written as

$$P(W \geq 3.5 \mid S = 1),$$

which denotes the probability that a smoking mother has a baby who weighs at least 3.5 kg.

Random variables can be classified based on their data type: categorical (nominal or ordinal) or numeric (discrete or continuous).

3 Discrete random variables

A categorical random variable has nominal or ordinal outcomes such as {red, blue, green} or {tiny, small, average, large, huge}.

A discrete random variable usually represents a count and has a countable number of possible values, such as {1, 2, 3, 4, 5, 6}; {0, 2, 4, 6, 8} or all integers.

A discrete or categorical random variable can be described by its **probability mass function (PMF)**.

The probability that the random variable X takes the value x is denoted $P(X = x) = p(x)$. Note that:

1. $0 \leq p(x) \leq 1$, a probability is always between 0 and 1.
2. $\sum_x p(x) = 1$, the probabilities over all possible outcomes sum to 1.

Example 3.1 (The number of dots on a die)**.** When rolling a die there are six possible outcomes: 1, 2, 3, 4, 5 and 6, each of which has the same probability, if the die is fair. The outcome of one die roll can be described by a random variable X. The probability of a particular outcome x is denoted $P(X = x)$ or $p(x)$.

The probability mass function of a fair six-sided die can be summarized in a table;

or in a bar plot;

Table 3.1: Probability mass function of a fair six-sided dice.

x	p(x)
1	0.17
2	0.17
3	0.17
4	0.17
5	0.17
6	0.17

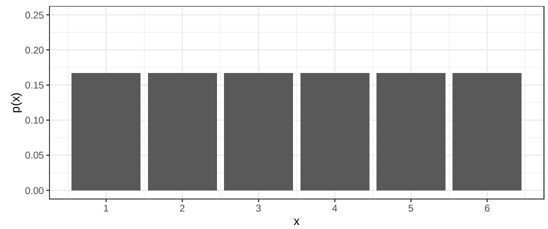


Figure 3.1: Probability mass function of a fair six-sided dice.

Table 3.2: Probability mass function of a nucleotide site.

x	p(x)
A	0.4
C	0.2
T	0.1
G	0.3

Example 3.2 (Nucleotide at a given site)**.** The nucleotide at a given genomic site can be one of the four nucleotides: {A, C, T, G}. Unlike the sides on a die, the four nucleotides are usually not equally likely. The nucleotide at the site can be described by a random variable, X. The probability mass function can be summarized in a table or a bar plot.

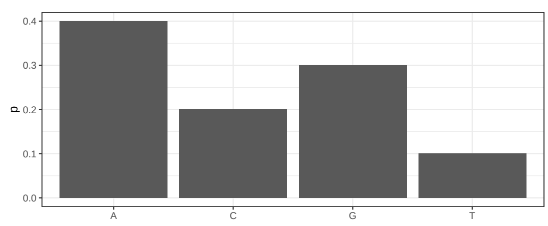


Figure 3.2: Probability mass function of a nucleotide site.

Example 3.3 (CFU)**.** The number of bacterial colony-forming units (CFU) on a plate is a random variable that can be described by a PMF such as the one shown in Figure 3.3.

The probability mass function can be used to compute the probability of an event. For example:

- Getting an even number when rolling a fair six-sided die: sum up the probability for 2, 4, and 6:
$$p(even) = p(2) + p(4) + p(6) = 0.5.$$

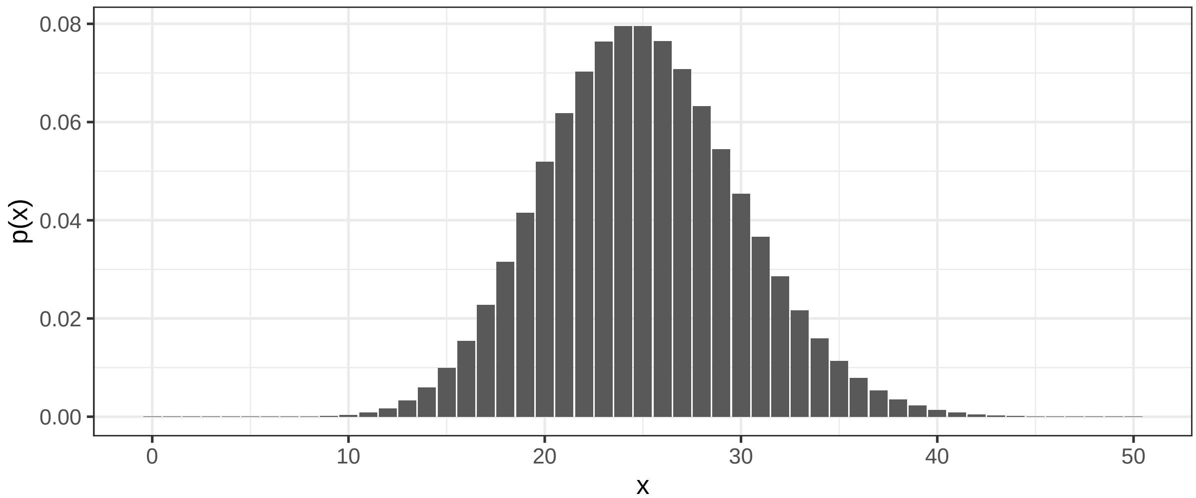


Figure 3.3: Probability mass distribution of the number of bacterial colonies on an agar plate.

- Getting fewer than 15 CFUs on an agar plate: sum up the probabilities:

$$P(X < 15) = \sum_{k=0}^{14} p(k) = 0.01.$$

The **cumulative distribution function (CDF)** is defined for discrete numeric random variables as

$$F(x) = P(X <= x).$$

In words, $F(x)$ is the probability that the random variable X takes a value less than or equal to x. For the CFU example, the probability of getting fewer than 15 CFUs on an agar plate can be written as $F(14) = P(X <= 14) = P(X < 15)$.

3.1 Expected value and variance

Two common properties of a probability distribution, such as a probability mass function, are the **expected value** and the **variance**.

The **expected value** is the average outcome of a random variable over many trials and is denoted $E[X]$ or μ and can be computed by summing up all possible outcome values weighted by their probability:

$$E[X] = \mu = \sum_{x \in S} xp(x),$$

where S is the set of all possible outcomes.

For a **uniform distribution**, where every outcome has the same probability, the expected value can be computed as the sum of all outcome values divided by the total number of outcome values.

For a finite population in which every object is equally likely to be selected, the expected value can also be computed as the population mean, i.e., by summing over outcome values for all objects in the population;

$$E[X] = \mu = \frac{1}{N}\sum_{i=1}^{N} x_i,$$

where N is the number of objects in the population.

The **variance** is a measure of spread defined as the expected value of the squared difference between the random variable and its expected value:

$$var(X) = \sigma^2 = E[(X - E[X])^2] = \sum_{x \in S}(x-\mu)^2 p(x).$$

The variance can also be computed by summing over the outcome value for every object in the population:

$$var(X) = \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2,$$

where N is the number of objects in the population.

The **standard deviation** is the square root of the **variance**. The standard deviation of a random variable is usually denoted σ. The standard deviation is always positive and on the same scale as the outcome values.

3.1.1 Linear transformations and combinations

The expected value and variance of combined or transformed random variables can often be computed using the following rules:

$$E(aX) = aE(X)$$

$$E(X + Y) = E(X) + E(Y)$$

$$E[aX + bY] = aE[X] + bE[Y],$$

where a and b are constants.

$$var(aX) = a^2 var(X)$$

For independent random variables X and Y:

$$var(aX + bY) = a^2 var(X) + b^2 var(Y).$$

3.2 Simulate distributions

Once a random variable's probability distribution is known, probabilities, such as $P(X = x), P(X < x)$ and $P(X \geq x)$, and properties, such as expected value and variance, of the random variable can be computed. If the distribution is not known, simulation can be used to approximate these values.

Example 3.4 (Simulate coin toss)**.** In a single coin toss, the probability of heads is 0.5. In 20 coin tosses, what is the probability of at least 15 heads?

The outcome of a single coin toss is a random variable, X, with two possible outcomes $\{H, T\}$. We know that $P(X = H) = 0.5$. The random variable of interest is the number of heads in 20 coin tosses, Y. The probability that we need to compute is $P(Y \geq 15)$.

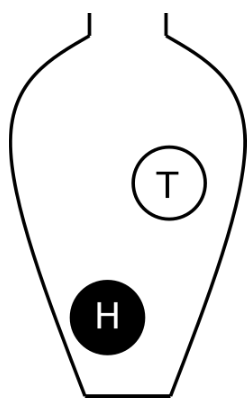


Figure 3.4: A coin toss. Urn model with one black ball (heads) and one white ball (tails).

A single coin toss can be modelled by an urn with two balls. When a ball is drawn randomly from the urn, the probability to get the black ball (heads) is $P(X = H) = 0.5$.

In R we can simulate random draws from an urn model using the function `sample`.

```r
set.seed(260602)
## A single coin toss
sample(c("H", "T"), size=1)
```

```
[1] "T"
```

```r
## Another coin toss
sample(c("H", "T"), size=1)
```

```
[1] "T"
```

Every time you run `sample`, a new coin toss is simulated.

If we want to simulate tossing 20 coins (or one coin 20 times), we can use the same urn model, provided that the ball is replaced after each draw.

The argument `size` tells the function how many balls we want to draw from the urn. To draw 20 balls from the urn, set `size=20`. Remember to replace the ball after each draw.

```r
## 20 independent coin tosses
(coins <- sample(c("H", "T"), size=20, replace=TRUE))
```

```
 [1] "T" "T" "T" "T" "H" "H" "T" "T" "H" "T" "T" "H" "T" "T" "H" "T" "T" "T" "H"
[20] "T"
```

How many heads did we get in the 20 random draws?

```r
## How many heads?
sum(coins == "H")
```

```
[1] 6
```

We can repeat this experiment (tossing 20 coins and counting the number of heads) several times to estimate the distribution of number of heads in 20 coin tosses.

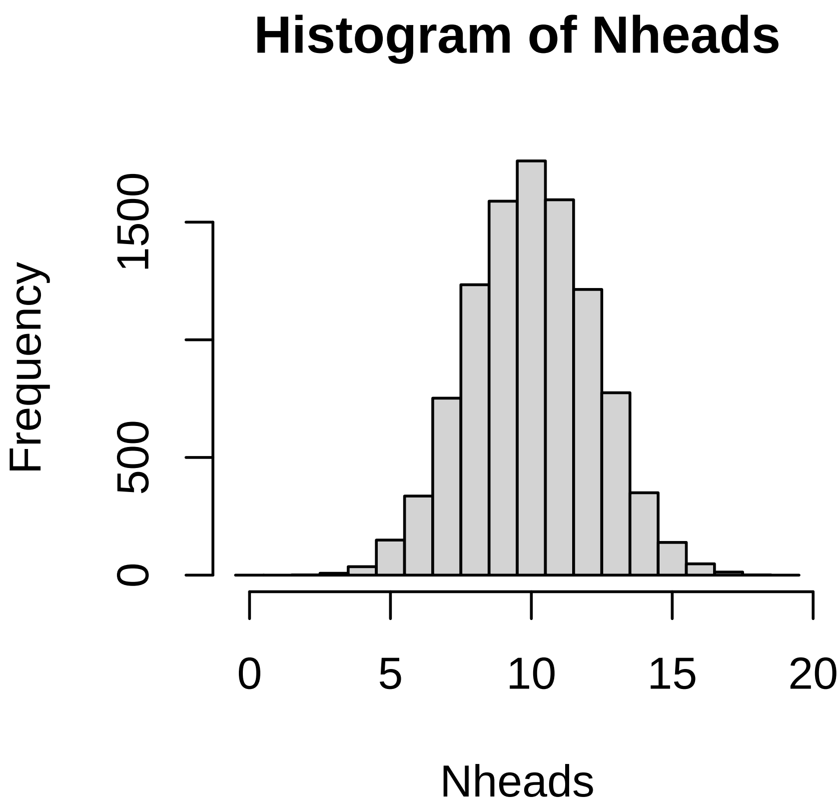
To do the same thing several times, we use the function `replicate`.

To simulate tossing 20 coins and counting the number of heads 10000 times, do the following:

```r
Nheads <- replicate(10000, {
  coins <- sample(c("H", "T"), size=20, replace=TRUE)
  sum(coins == "H")
})
```

Plot the distribution of the number of heads in a histogram.

```r
hist(Nheads, breaks=(0:20)-0.5)
```



Now, let's get back to the question: when tossing 20 coins, what is the probability of at least 15 heads?

$P(Y \geq 15)$

Count how many times out of our 10000 experiments, the number of heads is 15 or greater.

```r
sum(Nheads >= 15)
```

```
[1] 201
```

From this, we conclude that

$P(Y \geq 15) = 201/10000 = 0.0201$

Note that this is an approximation of the probability. The more times we repeat the experiment, the better our approximation will be. In fact, the number of heads in a series of independent coin tosses follows a binomial distribution, which is a discrete parametric distribution. We will get back to this later in the chapter.

Resampling can also be used to compute other properties of a random variable, such as its expected value.

The **law of large numbers** states that if the same experiment is performed many times, the average result will be close to the expected value.

The coin flip is a common example in statistics, but many situations with two outcomes can be modeled using similar models. If we consider the outcome of interest success (like heads in the coin example) and the alternative outcome failure, we can model, for example:

- Drug effect: A patient can respond to drug treatment (success) or not (failure)
- Side effect: After a treatment a patient might experience a side effect (success) or not (failure)
- Treatment or placebo: Randomly assign a study participant to the treatment or placebo group
- Antibiotic resistance: A bacterium is either resistant to an antibiotic or not

The probability of success and failure can be equal, like in the coin example, but they do not have to be. If the probability of a patient responding to a treatment is $p = 0.80$, then the probability of the patient not responding is $1 - p = 0.20$. This can be modeled using an urn model with 4 black balls (cured) and 1 white ball (not cured).

3.3 Parametric discrete distributions

A discrete parametric distribution is a probability distribution described by a set of parameters. In many situations, data can be assumed to follow a parametric distribution.

3.3.1 Uniform

In a uniform distribution, every possible outcome has the same probability. With n different outcomes, the probability for each outcome is $1/n$.

3.3.2 Bernoulli

A Bernoulli trial is a random experiment with two outcomes: success and failure. The probability of success, $P(success) = p$, is constant. The probability of failure is $P(failure) = 1 - p$.

When coding, it is convenient to code success as 1 and failure as 0.

The outcome of a Bernoulli trial is a discrete random variable, X.

$$P(X = x) = p(x) = \begin{cases} p & \text{if } x = 1 \text{ success} \\ 1 - p & \text{if } x = 0 \text{ failure} \end{cases}$$

Using the definitions of expected value and variance it can be shown that:

$$E[X] = p$$
$$var(X) = p(1 - p)$$

A Bernoulli trial can be simulated, as seen in previous section, using the function `sample`.

3.3.3 Binomial

A Bernoulli trial describes a single success/failure experiment. A binomial random variable counts the number of successes in n such trials.

The number of successes in a series of Bernouilli trials is a random variable, X, that follows the **binomial distribution**, if:

- The number of trials, n, is fixed.
- Each trial is independent.
- The probability of success, p, is the same for each trial.

$X = \sum_{i=1}^{n} Z_i$,

where all Z_i describe the outcome of independent and identical Bernoulli trials with probability p for *success* ($P(Z_i = 1) = p$).

The probability mass function of X is called the binomial distribution. In short, we use the notation:

$$X \sim Bin(n, p)$$

The probability mass function is

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

It can be shown that

$$E[X] = np$$
$$var(X) = np(1-p)$$

A binomial random variable is the number of successes when sampling n objects *with* replacement from an urn with objects of two types, of which the interesting type (success) has probability p.

The probability mass function, $P(X = k)$ can be computed using the R function `dbinom` and the cumulative distribution function $P(X \leq k)$ can be computed using `pbinom`.

Examples of binomial random variables;

- The number of patients responding to a treatment out of n patients in a study, if the probability of a patient responding to treatment is p.
- The number of patients experiencing a side effect out of n patients in a study, if the probability of a side effect is p.
- the number of mutations in a gene of length n, if the mutations are independent and identically distributed and the probability of a mutation at every single position is p.

3.3.4 Poisson

The Poisson distribution is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space, given that these events occur with a known constant mean rate and independently of the time since the last event. It is commonly used for rare events.

A rare disease has a very low probability for a single individual. The number of individuals in a large population that catch the disease in a certain time period can be modelled using the Poisson distribution.

The probability mass function has a single parameter, λ, the constant mean rate, and can be described as:

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}.$$

The parameter λ is both the expected value and the variance of the distribution:

In the case where the Poisson distribution is used to approximate the binomial distribution, the expected value $\lambda = n\pi$, where n is the number of objects sampled from the population and π is the probability of a single object being of the interesting type.

The variance is equal to the expected value:

$$var(X) = E[X] = \lambda = n\pi$$

The Poisson distribution can approximate the binomial distribution if n is large ($n > 20$) and π is small ($\pi < 0.05$ and $n\pi < 10$).

Examples of Poisson random variables;

- A rare disease has a very low probability for a single individual. The number of individuals in a large population that catch the disease in a certain time period is a Poisson random variable.
- Number of reads aligned to a gene region

3.3.5 Negative binomial

A negative binomial distribution describes the number of failures that occur before a specified number of successes, r, has occurred in a sequence of independent and identically distributed Bernoulli trials. r is also called the dispersion parameter.

In R: `dnbinom, pnbinom, qnbinom`

3.3.6 Geometric

The geometric distribution is a special case of the negative binomial distribution in which $r = 1$.

In R: `dgeom, pgeom, qgeom`

3.3.7 Hypergeometric distribution

The hypergeometric distribution occurs when sampling n objects *without* replacement from an urn with N objects of two types, of which there are K objects of the interesting type. The probability of selecting an object of the interesting type is $p = K/N$.

The probability mass function

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

can be computed in R using `dhyper` and the cumulative distribution function $P(X \leq k)$ can be computed using `phyper`.

Examples of hypergeometric random variables:

- In a student group of 25 individuals, 10 are R beginners. If 5 individuals are randomly chosen to belong to group A, the number of R beginners in group A is a hypergeometric random variable.
- A drug company is producing 1000 pills per day, 5% have an amount of active substance below an acceptable threshold. In a random sample of 10 pills, how many contain too little active substance? The number of pills with too little active substance is a hypergeometric random variable.
- You investigate the differential expression of 10000 genes, 150 of these genes belong to the super interesting pathway XXX. You run a black-box algorithm that reports 80 differentially expressed (DE) genes. If the black-box algorithm chooses DE genes at random, the number of DE genes that belong to pathway XXX is a hypergeometric random variable.

3.3.8 Summary of discrete distributions

- **Bernoulli**: one success/failure trial
- **Binomial**: number of successes in n independent Bernoulli trials
- **Poisson**: number of events in a fixed interval
- **Hypergeometric**: number of successes in sampling without replacement
- **Geometric**: number of failures before the first success
- **Negative binomial**: number of failures before the r-th success
- **Uniform**: all outcomes equally likely

The binomial, hypergeometric, negative binomial and Poisson distributions have similarities. For large N (large population), the hypergeometric and binomial distributions are very similar. The Poisson distribution has equal mean and variance, whereas the binomial has a variance smaller than the mean, and the negative binomial has a variance greater than the mean.

In R, probability mass functions, $P(X = x)$, for the binomial, hypergeometric, negative binomial and Poisson distributions can be computed using the functions `dbinom`, `dhyper`, `dnbinom` and `dpois`, respectively.

Cumulative distribution functions, $P(X \leq x)$ can be computed using `pbinom`, `phyper`, `pnbinom` and `ppois`.

Also, functions for computing an x such that $P(X \leq x) = q$, where q is a probability of interest, are available using `qbinom`, `qhyper`, `qnbinom` and `qpois`.

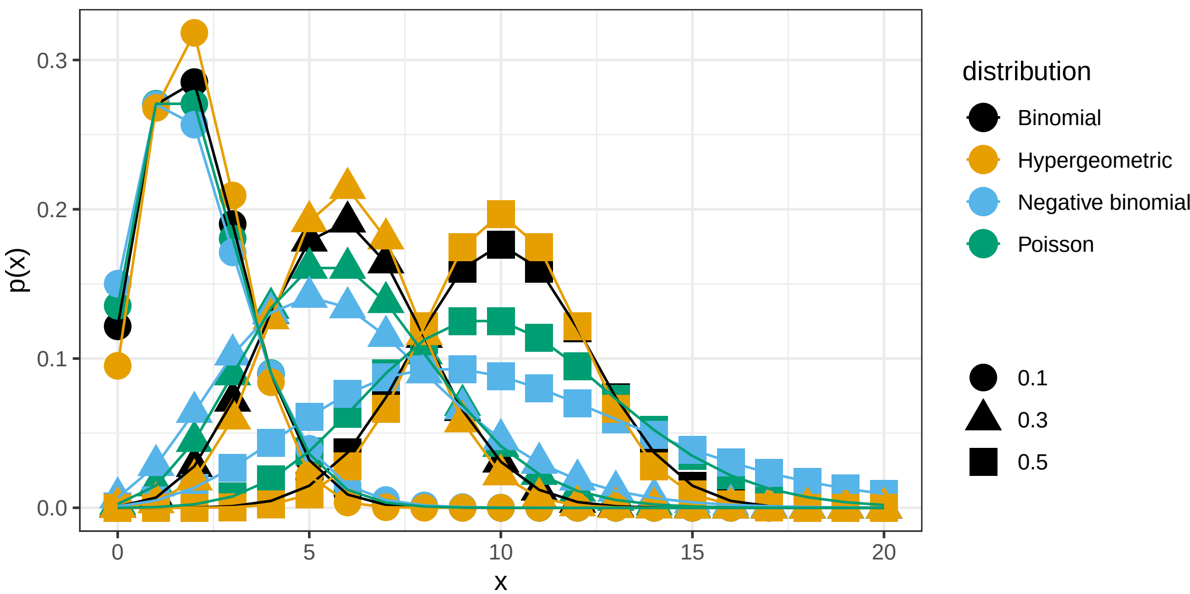


Figure 3.5: Probability mass functions for the binomial distribution (n=20, p=0.1, 0.3 or 0.5), hypergeometric distribution (N=100, n=20, p=0.1, 0.3 or 0.5), negative binomial distribution (n=20, r=n*p, p=0.1, 0.3 or 0.5) and Poisson distribution (n=20, p=0.1, 0.3 or 0.5).

Exercises: Discrete random variables

Introduction to probability

Exercise 3.1 (BRCA)**.** The probability of carrying one or more mutations in the breast cancer gene BRCA1 is 0.01. What is the probability of not carrying any mutations in BRCA1?

Hint

Use the rule of complement.

Solution

According to the rule of complement $P(no\,mutation) = 1 - P(mutations) = 1 - 0.01 = 0.99$

Exercise 3.2 (A coin toss)**.** When tossing a fair coin

a) what is the probability of heads?
b) what is the probability of tails?

Hint

Fair = equal probabilities

Solution

For a fair coin the probabilities of heads and tails are equal; $P(H) = P(T)$. According to the complement rule $P(H) = 1 - P(T)$.
It follows that

a) $P(H) = 0.5$
b) $P(T) = 0.5$

Exercise 3.3 (Number of children)**.** In a region in Sweden with many children the number of children per household is between 0 and 6. The probability mass function is as follows;

x	0	1	2	3	4	5	6
p(x)	0.14	0.20	0.27	0.19	0.13	0.05	0.02

In a randomly chosen household

a) what is the probability of exactly 3 children?
b) what is the probability of less than 3 children?
c) what is the probability of 3 or less children?
d) what is the probability of an even number of children?

In your answers, denote the probability with a mathematical expression (such as $P(X > 4)$) and calculate its value.

Solution

The number of children in a random household is a random variable. Let X (a random variable) denote the number of children in a household in the studied region.

a) $P(X = 3) = 0.19$
b) $P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = 0.14 + 0.20 + 0.27 = 0.61$
c) $P(X \leq 3) = P(X = 3) + P(X < 3) = 0.19 + 0.61 = 0.80$
d) $P(X \text{ is even}) = P(X = 0) + P(X = 2) + P(X = 4) + P(X = 6) = 0.14 + 0.27 + 0.13 + 0.02 = 0.56$

Exercise 3.4 (Rolling dice)**.** When tossing a fair six-sided die

a) what is the probability of getting 6?
b) what is the probability of an even number?
c) what is the probability of getting 3 or more?
d) what is the expected value of dots on the dice?

Hint

On a fair sided die, all six sides have equal probability.

Solution

The random variable, X, describes the number of dots on the upper face of a die.

a) $P(X = 6) = \frac{1}{6}$
b) $P(X \text{ is even}) = \frac{3}{6} = \frac{1}{2}$
c) $P(X \geq 3) = P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$

d) $E[X] = 1 * \frac{1}{6} + 2 * \frac{1}{6} + 3 * \frac{1}{6} + 4 * \frac{1}{6} + 5 * \frac{1}{6} + 6 * \frac{1}{6} = 3.5$

Simulation

Exercise 3.5 (Randomization)**.** In a clinical trial, enrolled patients are randomly assigned to treatment or control group with equal probability.

For a single patient, what is the probability of being assigned to

a) the treatment group?
b) the control group?

If 20 patients are enrolled in the study;

c) what is the probability of exactly 15 in the treatment group?
d) what is the probability of less than 7 in the treatment group?
e) What is the most probable number of patients in the treatment group?
f) what is the probability of 5 or fewer patients in the control group?
g) what is the probability of 2 or fewer patients in the treatment group?

Answer

a. $P(T) = 0.5$
b. $P(C) = 0.5$

Solution

a. The probability of assigning to control (C) and treatment (T) group are equal and sum up to 1. Hence $P(T) = 0.5$ and;

b. $P(C) = 0.5$

Simulate the assignment of patients into T or C groups.

```
## Randomization for a single patient
sample(c("T", "C"), size=1)
```

```
[1] "C"
```

```
## Randomize 20 independent patients
(patients <- sample(c("T", "C"), size=20, replace=TRUE))
```

```
 [1] "C" "T" "C" "C" "T" "T" "T" "C" "T" "C" "C" "T" "T" "T" "C" "T" "T" "C" "T"
[20] "C"
```

```r
## How many patients are assigned to treatment group?
sum(patients == "T")
```

```
[1] 11
```

```r
## Simulate by repeating 10000 times
Ntreat <- replicate(10000, {
  patients <- sample(c("T", "C"), size=20, replace=TRUE)
  sum(patients == "T")
})
```

c. Probability of exactly 15 T

```r
## Proportion of the 10000 repeats with exactly 15 T
mean(Ntreat==15)
```

```
[1] 0.015
```

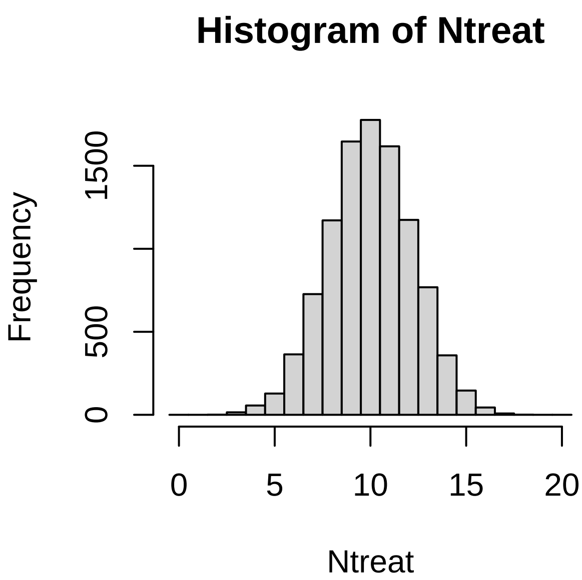
d. Probability of less than 7 T

```r
mean(Ntreat<7)
```

```
[1] 0.056
```

e. What is the most probable number of T patients?

```r
## plot the distribution and read the graph
hist(Ntreat, breaks=0:21-0.5)
```




```
## or tabulate
table(Ntreat)
```

```
Ntreat
   2    3    4    5    6    7    8    9   10   11   12   13   14   15   16   17
   1   15   56  128  364  727 1171 1646 1776 1617 1174  768  358  146   44    8
  18
   1
```

f. What is the probability of 5 C or fewer?

To get five or fewer C out of 20 throws is equal to getting 15 or more T out of 20.

```
## probability of 15 T or more
mean(Ntreat>=15)
```

```
[1] 0.02
```

g. what is the probability of 2 T or fewer?

```
mean(Ntreat<=2)
```

```
[1] 1e-04
```

```
sum(Ntreat<=2)
```

```
[1] 1
```

```r
## with this low number of observations, more repeats is required to get a more accurate an

Ntreat <- replicate(1000000, {
  patients <- sample(c("T", "C"), size=20, replace=TRUE)
  sum(patients == "T")
})
sum(Ntreat<=2)
```

```
[1] 202
```

```r
mean(Ntreat<=2)
```

```
[1] 2e-04
```

Exercise 3.6 (Bacterial colonies)**.** In a bacterial sample, 1/6 are antibiotic resistant. From bacterial colonies on an agar plate, you randomly pick 10 colonies and investigate how many that are antibiotic resistant.

a) Define the random variable of interest
b) What are the possible outcomes?
c) Using simulation, estimate the probability mass function
d) what is the probability to get at least 5 antibiotic resistant colonies?
e) Which is the most likely number of antibiotic resistant colonies?
f) What is the probability to get exactly 2 antibiotic resistant colonies?
g) On average how many antibiotic resistant colonies would you get if the experiment is repeated many times?

Hint

Think of the dice example, where the probability of getting 'six' on one die is 1/6.

Solution

a. X, the number of antibiotic resistant colonies out of 10.

b. $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$

c.

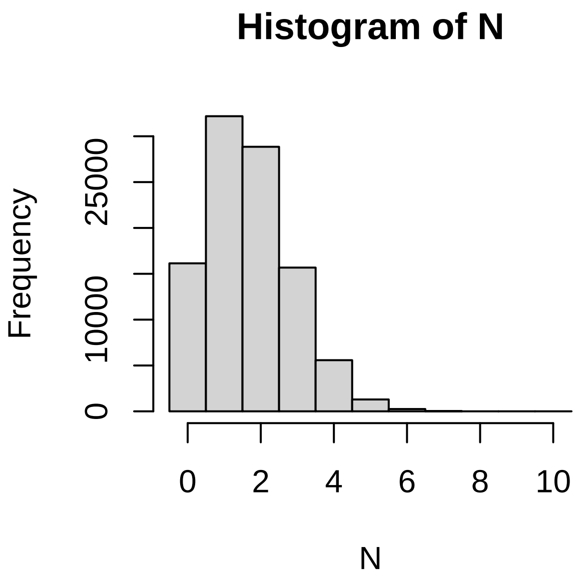
```r
##Simulate picking 10 colonies and counting the number of antibiotic resistant ones.
N <- replicate(100000, sum(sample(1:6, size=10, replace=TRUE)==6))
table(N)
```

```
N
    0     1     2     3     4     5     6     7     8
16142 32185 28850 15676  5577  1294   249    26     1
```

```r
##The probability mass function
table(N)/length(N)
```

```
N
      0       1       2       3       4       5       6       7       8
0.16142 0.32185 0.28850 0.15676 0.05577 0.01294 0.00249 0.00026 0.00001
```

```r
hist(N, breaks=(0:11)-0.5)
```



d) 0.015

```
[1] 1570
```

```
[1] 0.016
```

```
[1] 0.016
```

e) 1 (Use the PMF to answer this question)

f) 0.29

```
mean(N==2)
```

```
[1] 0.29
```

g) The law of large numbers states that if the same experiment is performed many times the average of the result will be close to the expected value.

1.7

```
mean(N)
```

```
[1] 1.7
```

```
10*1/6
```

```
[1] 1.7
```

Exercise 3.7 (Pollen allergy)**.**

a. 30% of a large population is allergic to pollen. If you randomly select 3 people to participate in your study, what is the probability that none of them will be allergic to pollen?

Solution

```
## Solution using 100 replicates
x <- replicate(100, sum(sample(c(0,0,0,0,0,0,0,1,1,1), size=3, replace=TRUE)))
table(x)
```

```
x
 0  1  2  3 
36 39 23  2 
```

```
mean(x==0)
```

```
[1] 0.36
```

```
## Solution using 1000 replicates
x <- replicate(1000, sum(sample(c(0,0,0,0,0,0,0,1,1,1), size=3, replace=TRUE)))
table(x)
```

```
x
  0   1   2   3 
347 426 199  28 
```

```r
mean(x==0)
```

```
[1] 0.35
```

```r
## Solution using 100000 replicates
x <- replicate(100000, sum(sample(c(0,0,0,0,0,0,0,1,1,1), size=3, replace=TRUE)))
table(x)
```

```
x
    0     1     2     3 
34118 44333 18862  2687 
```

```r
mean(x==0)
```

```
[1] 0.34
```

b. In a class of 20 students, 6 are allergic to pollen. If you randomly select 3 of the students to participate in your study, what is the probability that none of them will be allergic to pollen?

Solution

```r
## Solution using 100000 replicates
x <- replicate(100000, sum(sample(rep(c(0, 1), c(14, 6)), size=3, replace=FALSE)))
table(x)
```

```
x
    0     1     2     3 
31861 48043 18337  1759 
```

```r
mean(x==0)
```

```
[1] 0.32
```

c. Of the 200 persons working at a company, 60 are allergic to pollen. If you randomly select 3 people to participate in your study, what is the probability that none of them are allergic to pollen?

Solution

```r
## Solution using 100000 replicates
x <- replicate(100000, sum(sample(rep(c(0, 1), c(140, 60)), size=3, replace=FALSE)))
table(x)
```

```
x
    0     1     2     3 
34403 44351 18647  2599 
```

```r
mean(x==0)
```

```
[1] 0.34
```

d. Compare your results in a, b and c. Did you get the same results? Why/why not?

Solution

The results differ. In (a), the probability of selecting an allergic person is constant because sampling is with replacement (or from a very large population modeled that way). In (b) and (c), sampling is without replacement, so the probability changes after each selection.

Parametric discrete distribution

Exercise 3.8 (Pollen)**.** Do Exercise 3.7 again, but using parametric distributions. Compare your results.

Hint

Use the binomial distribution in (a) and the hypergeometric distribution in (b) and (c).

Solution

```r
## 1.6 Solution using the Binomial distribution
pbinom(0, 3, 0.3)
```

```
[1] 0.34
```

```r
## 1.7 Solution using the hypergeometric distribution
phyper(0, 6, 20-6, 3)
```

```
[1] 0.32
```

```r
## 1.8 Solution using the hypergeometric distribution
phyper(0, 60, 200-60, 3)
```

```
[1] 0.34
```

Exercise 3.9 (Gene set enrichment analysis)**.** You have analyzed 20000 genes and a bioinformatician you are collaborating with has sent you a list of 1000 genes that she says are important. You are interested in a particular pathway A. 200 genes in pathway A are represented among the 20000 genes, 20 of these are in the bioinformaticians important list.

If the bioinformatician selected the 1000 genes at random, what is the probability to see 20 or more genes from pathway A in this list?

Solution

200 A genes.
20000-200 non-A genes.
1000 genes selected.

```r
## 200 A genes.
## 20000-200 non-A genes.
## 1000 genes selected.
## P(X>=20) = 1 - P(X<=19)
## lower.tail gives the complement
phyper(19, 200, 20000-200, 1000, lower.tail=FALSE)
```

```
[1] 0.0025
```

Exercise 3.10 (Chance of meeting boss)**.** Your boss comes in to the office three days per week. You do also come in to work three days per week. If you both choose which days to come in to work at random, what is the probability that a particular week you are in the office at the same time 0, 1, 2 or 3 days, respectively?

Solution

```
## Assume there are 5 workdays in a week.
x <- replicate(100000, {x<-sample(1:5, 3); y<-sample(1:5,3); length(intersect(x,y))})
table(x)
```

```
x
    1     2     3 
29998 60130  9872 
```

```
dhyper(0:3, 3, 2, 3)
```

```
[1] 0.0 0.3 0.6 0.1
```

Exercise 3.11 (Rare disease)**.** A rare disease affects 3 in 100000 in a large population. If 10000 people are randomly selected from the population, what is the probability

a) that no one in the sample is affected?
b) that at least two in the sample are affected?

Solution

a)

```
n <- 10000
p <- 3/100000
ppois(0, n*p)
```

```
[1] 0.74
```

b)

```
ppois(1, n*p, lower.tail=FALSE)
```

```
[1] 0.037
```

4 Continuous random variables

A continuous random variable is not limited to discrete values, but can take any value within one or several intervals on the real line.

Examples of continuous random variables include weight, height, speed, and fluorescence intensity.

A continuous random variable can be described by its **probability density function**, PDF. For a continuous random variable, the probability of a single value is zero, i.e., $P(X = x) = 0$. Probabilities are instead represented by the area under the probability density function.

For discrete random variables, probabilities are assigned to individual values through a probability mass function (PMF). For continuous random variables, probabilities are assigned to intervals through a probability density function (PDF).

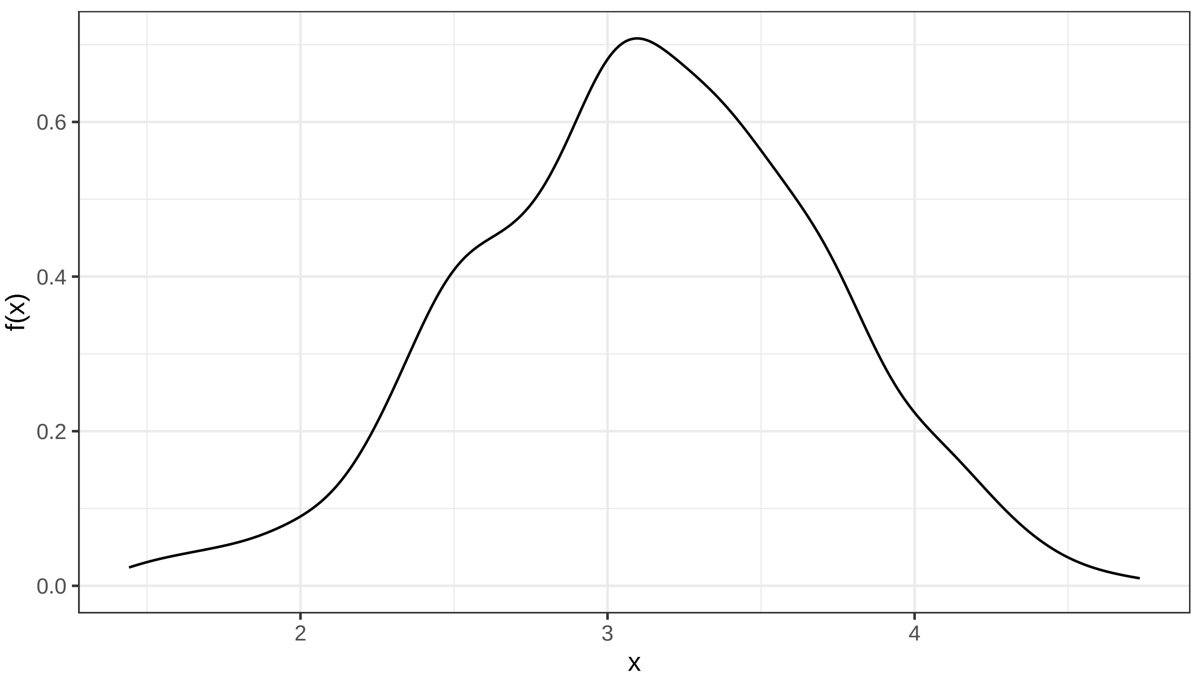


Figure 4.1: Probability density function of the weight of a newborn baby.

The probability density function, $f(x)$, is defined such that the total area under the curve is 1.

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

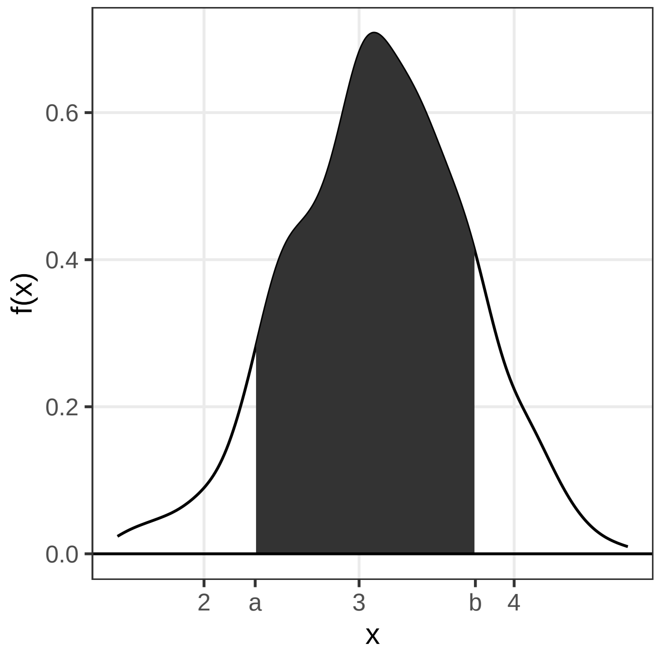


Figure 4.2: The probability $P(a \leq X \leq b)$ can be computed by computing the area under the probability density function between a and b.

Probabilities are computed by integrating the probability density function over the interval of interest. The area under the curve from a to b is the probability that the random variable X takes a value between a and b:

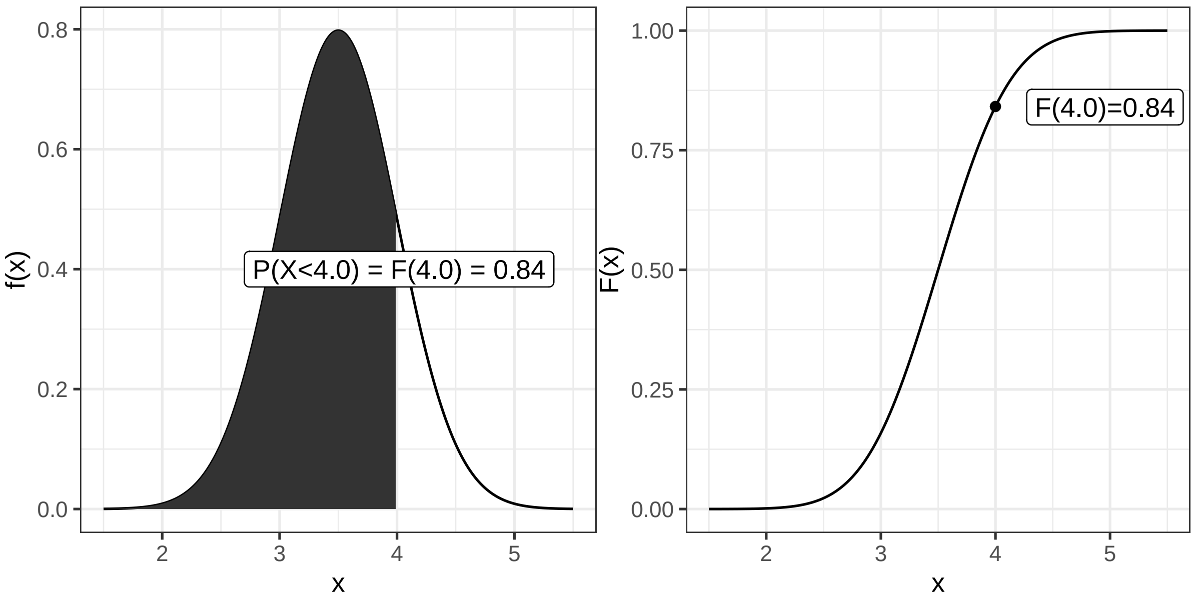
$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

The **cumulative distribution function**, CDF, sometimes called just the distribution function, $F(x)$, is defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$$

The cumulative distribution function is increasing from 0 to 1.

$$P(X \leq x) = F(x)$$



(a) PDF (b) CDF

Figure 4.3: Probability density function (PDF) and cumulative distribution function (CDF).

As we know that the total probability (over all x) is 1, we can conclude that

$$P(X > x) = 1 - F(x)$$

and thus

$$P(a \leq X \leq b) = F(b) - F(a)$$

Note that $P(a \leq X \leq b) = P(a < X < b) = P(a < X \leq b) = P(a \leq X < b)$, as the probability of a single value is zero.

4.1 Parametric continuous distributions

Two important properties of a distribution are its expected value, μ, which describes the distribution's location, and its variance, σ^2, which describes the spread.

The expected value, or population mean, is defined as;

$$E[X] = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

We will learn more about the expected value and how to estimate a population mean from a sample later in the course.

The population variance is defined as the expected value of the squared distance from the population mean;

$$\sigma^2 = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx$$

These definitions are similar to the definitions of expected value and variance for discrete random variables, but with summation replaced by integration.

The square root of the variance is the standard deviation, σ.

4.2 Normal distribution

The normal distribution (sometimes referred to as the Gaussian distribution) is a common bell-shaped probability distribution. Many continuous random variables can be described by the normal distribution or be approximated by the normal distribution. The normal distribution is especially important because many statistical methods are based on it directly, and many other distributions can be approximated by it.

The normal probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

describes the distribution of a normal random variable, X, with expected value μ and standard deviation σ, e and π are two common mathematical constants, $e \approx 2.71828$ and $\pi \approx 3.14159$.

In short, we write $X \sim N(\mu, \sigma^2)$ to denote that X is normally distributed with expected value μ and variance σ^2 (and standard deviation σ).

The normal distribution is symmetric around μ, and its density $f(x) \to 0$ as $x \to \infty$ and as $x \to -\infty$.

As $f(x)$ is known, the cumulative distribution function $F(x) = \int_{-\infty}^{x} f(x)dx$ can be computed.

Using transformation rules we can define Z, a random variable that is standard normally distributed (mean 0 and standard deviation 1).

$$Z = \frac{X-\mu}{\sigma},\ Z \sim N(0,1)$$

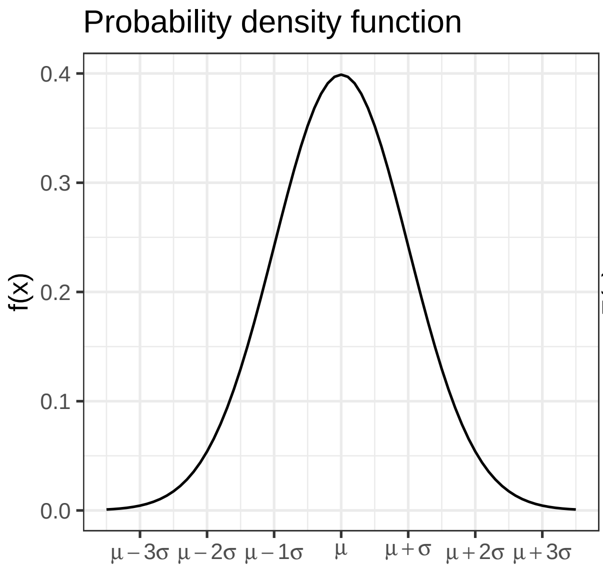


Figure 4.4: Normal probability density function and cumulative distribution functions.

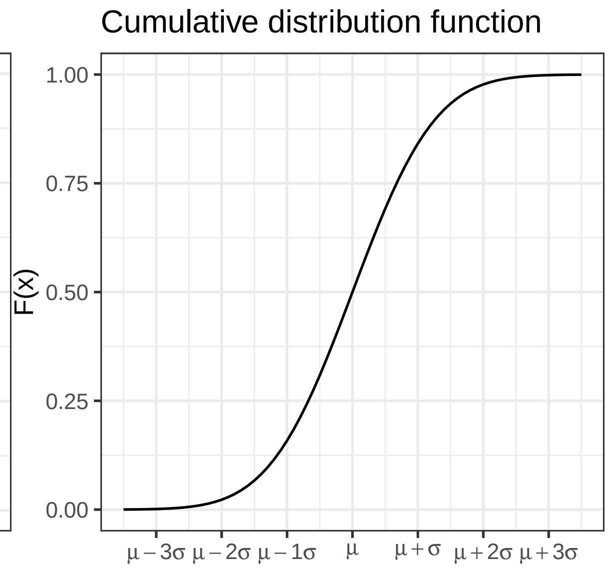


Figure 4.5: Normal probability density function and cumulative distribution functions.

Standardization transforms a normal random variable X to a standard normal random variable Z, which makes probability calculations easy using tables and software. The cumulative standard normal distribution is denoted $F(z)$ and can be computed in R using the function `pnorm(z)`.

Table 4.1: Cumulative distribution function for the standard normal distribution. The table gives $F(z) = P(Z \leq z)$ for standard normal Z.

	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621

1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Some values of particular interest:

$$F(1.64) = 0.95$$
$$F(1.96) = 0.975$$

Because the normal distribution is symmetric,

$$F(-z) = 1 - F(z).$$

$$F(-1.64) = 0.05$$
$$F(-1.96) = 0.025$$
$$P(-1.96 < Z < 1.96) = 0.95$$

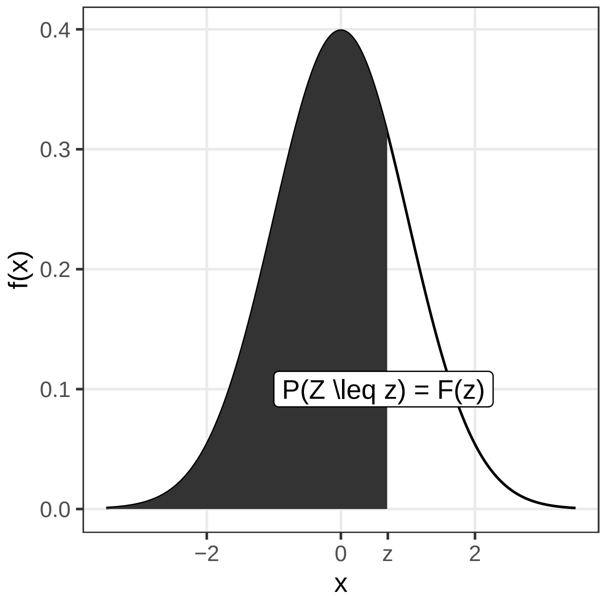


Figure 4.6: The shaded area under the curve is the tabulated value $P(Z \leq z) = F(z)$.

4.2.1 Sum of two normal random variables

If $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ are two independent normal random variables, then their sum is also a random variable:

$$X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

and

$$X - Y \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$$

The standard deviation of the sum or difference of two independent normal random variables is thus $\sqrt{\sigma_1^2 + \sigma_2^2}$.

This can be extended to the case with n independent and identically distributed random variables X_i $(i = 1 \dots n)$. If all X_i are normally distributed with mean μ and standard deviation σ,

$$X_i \sim N(\mu, \sigma^2)$$

then

$$\sum_{i=1}^{n} X_i \sim N(n\mu, n\sigma^2)$$

and therefore

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

4.3 Central limit theorem, CLT

The **central limit theorem** states that the sum or mean of n independent and identically distributed random variables is approximately normally distributed, if n is large enough. The mean and variance of the sum are $n\mu$ and $n\sigma^2$, respectively.

As a result of the central limit theorem, sample means and sample proportions are often approximately normally distributed, when the sample is large enough. A rule of thumb is that the sample size $n > 30$ is large enough, but this of course depends on the distribution of the individual random variables.

Example 4.1 (A skewed distribution)**.** A left skewed distribution has a heavier left tail than right tail. An example might be age at death of natural causes, as few individuals die of natural causes at young ages.

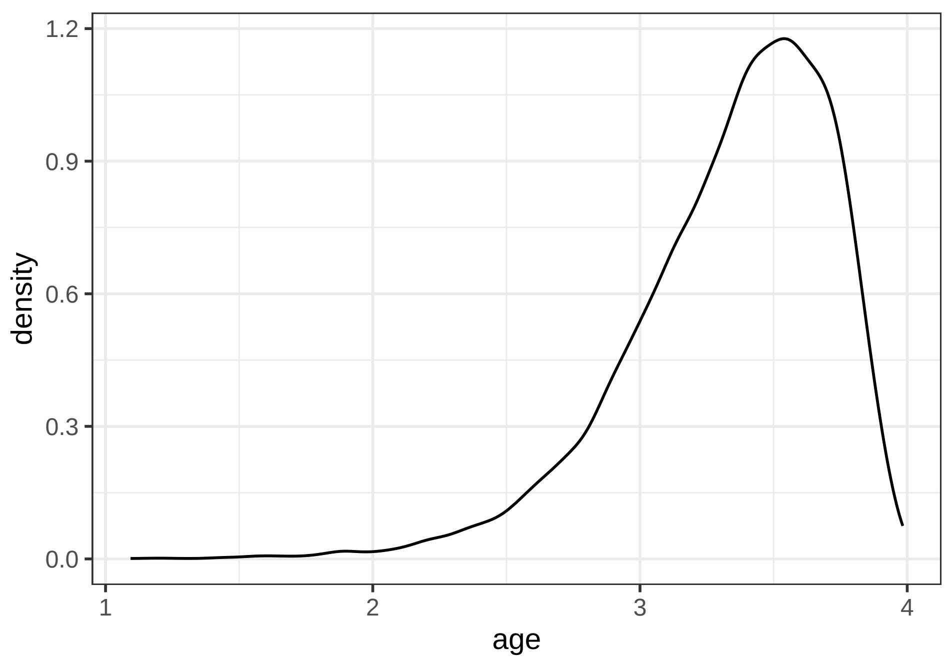


Figure 4.7: A left skewed distribution. Can for example show the distribution of age of a mouse who died of natural causes.

Randomly sample 3, 5, 10, 15, 20, 30 values and compute the mean value, m. Repeat many times to get the distribution of mean values.

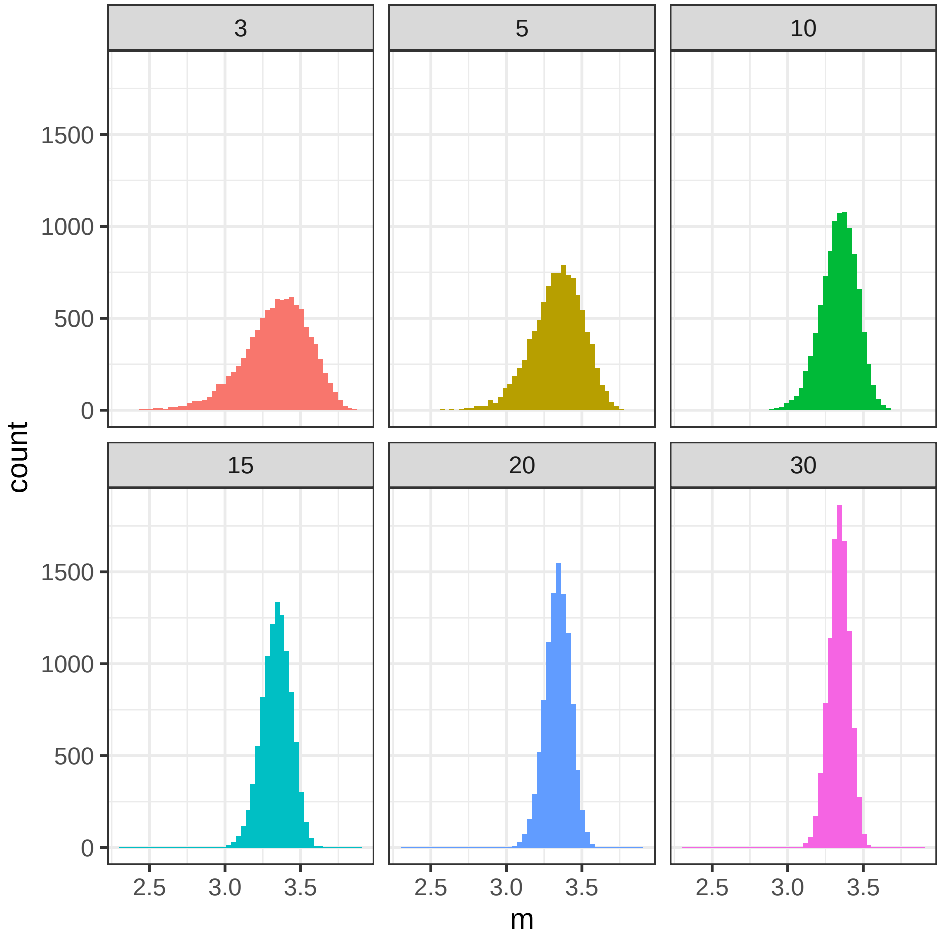


Figure 4.8: Distribution of sample means, where the means are computed based on random samples of sizes 3, 5, 10, 15, 20 and 30, respectively.

Note, mean is just the sum divided by the number of samples n.

As the sample size increases, the distribution of the sample mean becomes more symmetric and closer to normal.

In addition to the normal distribution, three other continuous distributions are especially important in statistical inference: the χ^2, F, and t distributions

4.4 χ^2-distribution

The random variable $Y = \sum_{i=1}^{n} X_i^2$ is χ^2 distributed with n degrees of freedom, if X_i are independent identically distributed random variables $X_i \in N(0,1)$.

In short $Y \in \chi^2(n)$.

Example 4.2. The sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ is such that $\frac{(n-1)S^2}{\sigma^2}$ is χ^2 distributed with $n-1$ degrees of freedom.

4.5 F-distribution

If $U \sim \chi^2(n_1)$ and $V \sim \chi^2(n_2)$ are independent, then the ratio

$$\frac{U/n_1}{V/n_2} \sim F(n_1, n_2)$$

is F-distributed with n_1 and n_2 degrees of freedom.

```
Warning: Removed 47 rows containing missing values or values outside the scale range
(`geom_line()`).
```

Example 4.3. The ratio of two sample variances is F-distributed, if the two samples are independent and normally distributed.

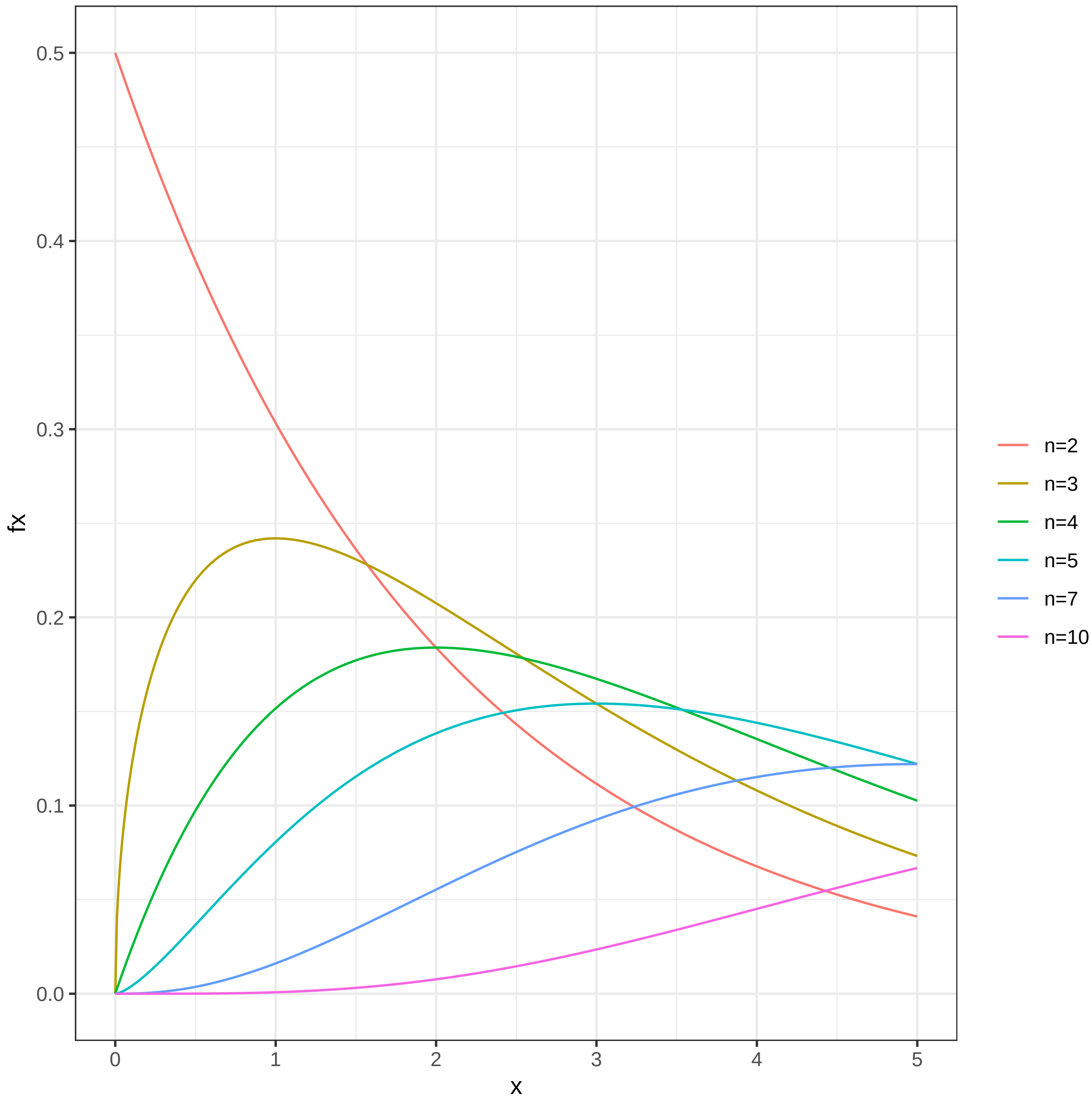


Figure 4.9: The χ^2-distribution.

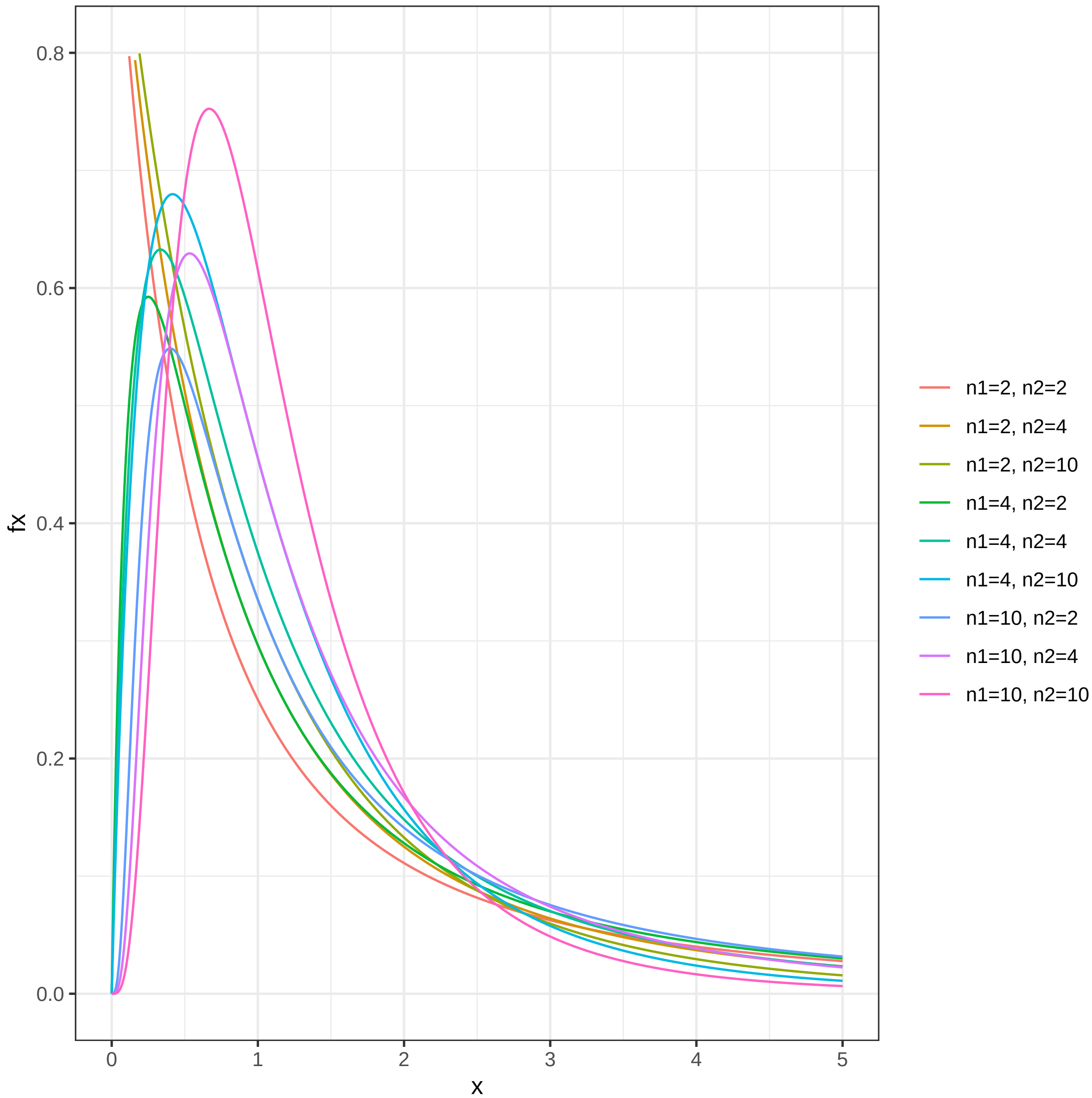


Figure 4.10: The F-distribution

4.6 t-distribution

If $Z \sim N(0,1)$ and $U \sim \chi^2(v)$ are independent, then the random variable

$$T = \frac{Z}{\sqrt{U/v}} \sim t(v)$$

is t-distributed with v degrees of freedom.

A common example is based on the sample mean, $\bar{X}$, and sample variance, S^2, of a sample of size n from a normal distribtion. The standardized sample mean

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

is t-distributed with $n-1$ degrees of freedom.

4.7 Distributions in R

Probability density functions for the normal, t, χ^2 and F distributions can in R be computed using functions `dnorm`, `dt`, `dchisq`, and `df`, respectively.

Cumulative distribution functions can be computed using `pnorm`, `pt`, `pchisq` and `pf`.

Also, functions for computing an x such that $P(X < x) = q$, where q is a probability of interest are available using `qnorm`, `qt`, `qchisq` and `qf`.

As with distributions, R uses the general naming scheme `r<dist>`, `p<dist>`, `q<dist>` and `d<dist>` for random number generation, cumulative distribution function, quantile function and probability density function, respectively.

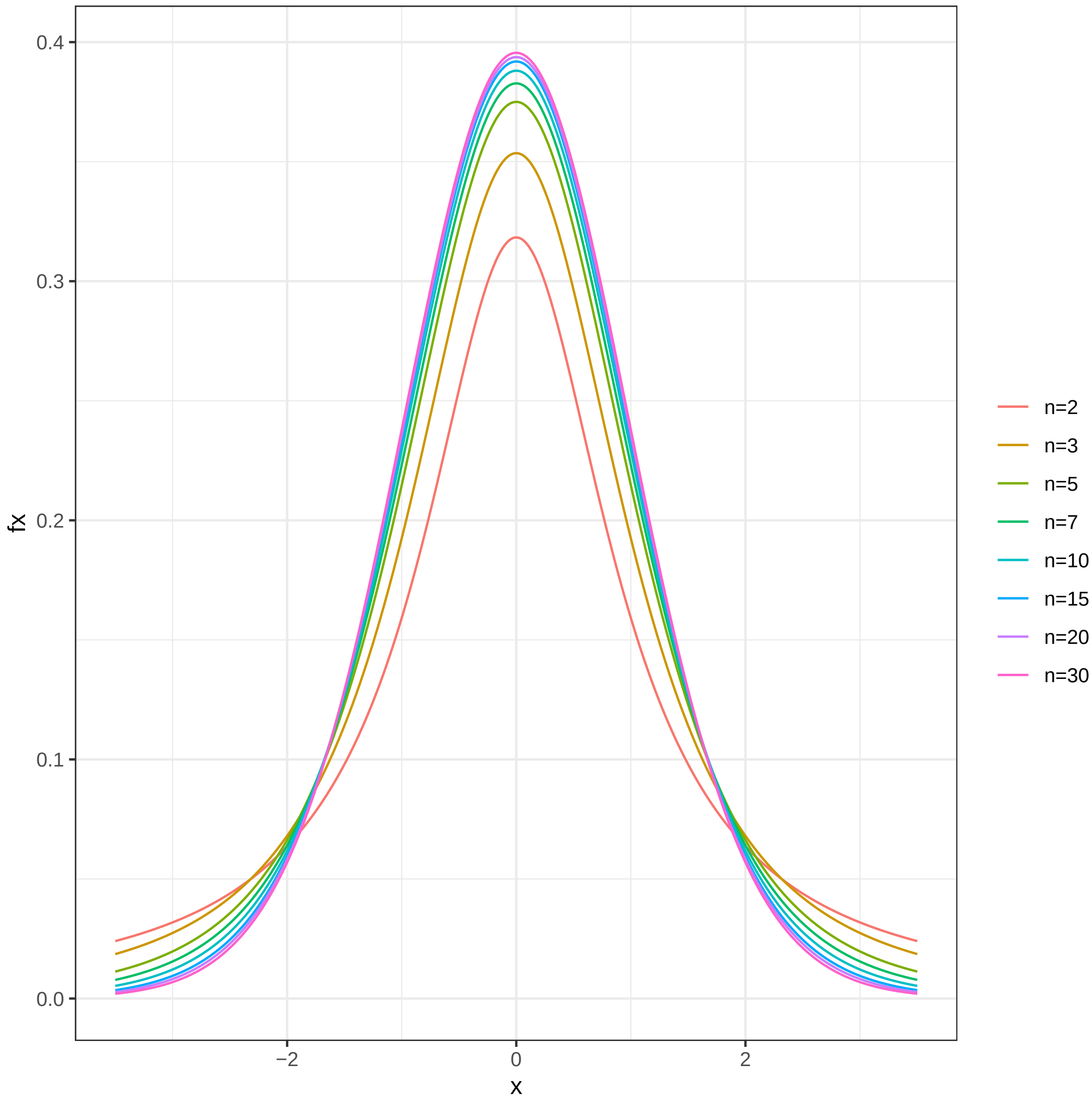


Figure 4.11: The t-distribution.

5 Sampling and experimental design

5.1 Random sampling

In many (most) experiments it is not feasible (or even possible) to examine the entire population. Instead a random sample is studied.

A **random sample** is a random subset of individuals from a population.

There are different techniques for performing random sampling, two common techniques are **simple random sampling** and **stratified random sampling**.

A **simple random sample** is a random subset of individuals from a population, where every individual has the same probability of being chosen. Simple random sampling can be performed using the urn model. If every individual in the population is represented by one ball, a simple random sample of size n can be achieved by drawing n balls from the urn, without replacement.

In **stratified random sampling** the population is first divided into subpopulations based on important attributes, e.g. sex (male/female), age (young/middle aged/old) or BMI (underweight/normal weight/overweight/obese). Simple random sampling is then performed within each subpopulation.

Stratified sampling is often used to ensure that the sample is representative of the population with respect to the stratifying variable and may reduce the variability of the sample estimate, if the strata are more homogeneous than the population as a whole.

5.2 Principles of experimental design

Random sampling concerns how to select a sample from a population, whereas experimental design concerns how to assign experimental units to treatments (or other experimental conditions) in an experiment.

When you plan your experiment and assign experimental units to treatment/control group (or similar), it is important to consider extraneous variables, i.e. variables that are not your main interest but that might affect the studied experimental outcome or the variable of interest. These variables can be properties of the experimental units such as age, sex etc. but also variables introduced in the experiment, like batch, experiment date, laboratory personell etc. If such variables are not controlled for, they may confound the relationship between the treatment and outcome, and lead to incorrect conclusions.

Fundamental to experimental design are the three principles; **replication**, **randomization** and **blocking**.

Replication. Replication is the repetition of the same experiment, with the same conditions. **Biological replicates** are measurements of different biological units under the same conditions, whereas **technical replicates** are repeated measurements of the same biological unit under the same conditions.

Technical replicates improve the precision of the measurement, but they do not increase the number of independent biological observations.

Randomization. Experimental units are not identical, hence by assigning experimental units to treatment/control at random we can avoid unnecessary bias. It is also important to perform the measurements in random order.

Blocking. Blocking is grouping experimental units into blocks consisting of units that are similar to one another and assigning units within a block to treatment/control at random. Blocking reduces known but irrelevant sources of variation between units and thus allows greater precision in the estimation of the source of variation under study.

The experimental units can be grouped into blocks according to their properies (e.g. age, sex, etc). Units within a block will be more similar than between blocks. By assigning units within a block to treatment/control at random, the variation due to differences between blocks (that are not relevant to the studied outcome) can be reduced.

In many retrospective studies it is not possible to assign patients into treatment/control or sick/healthy. Experimental design is still important for controling sources of variation introduced during the experiment.

Block what you can; randomize what you cannot.

Once a sample has been collected, we can summarize it using sample statistics such as proportions, means and variances.

5.3 Sample statistics and sampling distributions

Summary statistics can be computed for a sample, such as the sum, proportion, mean and variance.

Notation: A random sample consists of observations $x_1, x_2, \dots, x_n$ of independent and identically distributed random variables $X_1, X_2, \dots, X_n$ all with the probability distribution D.

5.3.1 Sample proportion

The proportion of a population with a particular property, the **population proportion**, is denoted π.

The number of individuals with the property in a simple random sample of size n is a random variable X. The proportion of individuals in a sample with the property is also a random variable;

$$P = \frac{X}{n}$$

with expected value

$$E[P] = \frac{E[X]}{n} = \frac{n\pi}{n} = \pi$$

.

The sample proportion, P, is said to be an **unbiased** estimator of the **population proportion**, as it's expected value is the population proportion π.

5.3.2 Sample mean

For a particular sample of size n; $x_1, \ldots, x_n$, the sample mean is denoted $\bar{x}$. The sample mean is calculated as;

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Note that the mean of n independent identically distributed random variables, X_i, is itself a random variable;

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

If X_i are normaly distributed $X_i \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

When we only have a sample of size n, the sample mean $barx$ is our best estimate of the population mean. It is possible to show that the sample mean is an **unbiased** estimator of the population mean, i.e. the average (over many size n samples) of the sample mean is μ.

$$E[\bar{X}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n} n\mu = E[X] = \mu$$

5.3.3 Sample variance

The sample variance is computed as;

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The denominator $n-1$ makes the sample variance an unbiased estimate of the population variance.

5.3.4 Sampling distribution

A sampling distribution is a probability distribution of a sample property. A sampling distribution is obtained by sampling many times from the studied population.

5.3.5 Standard error

Eventhough the sample can be used to calculate unbiased estimates of the population property, the sample estimate will not be perfect. The standard deviation of the sampling distribution is called the **standard error**, SE. Different statistics have different standard errors, for example the standard error of the mean is called the **standard error of the mean**, SEM.

For the sample mean, $\bar{X}$, the variance is

$$var(\bar{X}) = E[(\bar{X} - \mu)^2]$$

If the observations are independent, then

$$var(\bar{X}) = var(\frac{1}{n}\sum_i X_i) = \frac{1}{n^2}\sum_i var(X_i) = \frac{1}{n^2} nvar(X) = \frac{\sigma^2}{n}$$

The standard error of the mean is thus;

$$SEM = \frac{\sigma}{\sqrt{n}}$$

Replacing σ with the sample standard deviation, s, we get an estimate of the standard error of the mean;

$$SEM \approx \frac{s}{\sqrt{n}}$$

Exercises: Continuous random variables

Parametric continuous distribution

Exercise 5.1 (The normal table)**.** Let $Z \sim N(0,1)$ be a standard normal random variable, and compute;

a. $P(Z < 1.64)$
b. $P(Z > -1.64)$
c. $P(-1.96 < Z)$
d. $P(Z < 2.36)$
e. An a such that $P(Z < a) = 0.95$
f. A b such that $P(Z > b) = 0.975$

Note, this exercise can be solved using the standard normal table or using the R functions `pnorm`and `qnorm`.

Solution

a. $P(Z < 1.64) = P(Z \leq 1.64) =$ [from table] $= 0.9495$
b. $P(Z > -1.64) =$ [symmetry] $= P(Z < 1.64) = 0.9495$
c. $P(-1.96 < Z) =$ [symmetry] $= P(Z < 1.96) =$ [from table] $= 0.975$
d. $P(Z < 2.36) =$ [from table] $= 0.9909$
e. Standard normal table gives that $a = 1.64$
f. $P(Z > b) = 0.975$, symmetry gives that $P(Z < -b) = 0.975$. Look-up in the standard normal table gives that $-b = 1.96$, hence $b = -1.96$.

Exercise 5.2 (Exercise in standardization/transformation)**.** If $X \sim N(3,4)$, compute the probabilities

a. $P(X < 5)$
b. $P(3 < X < 5)$
c. $P(X \geq 7)$

Solution

Mean $\mu = 3$ and variance $\sigma^2 = 4$, hence standard deviation $\sigma = 2$.

a. $P(X < 5) = P\left(\frac{X-3}{2} < \frac{5-3}{2}\right) = P(Z < 1) = \text{[from table]} = 0.8413$
b. $P(3 < X < 5) = P\left(\frac{3-3}{2} < \frac{X-3}{2} < \frac{5-3}{2}\right) = P(0 < Z < 1) = P(Z < 1) - P(Z < 0) = 0.8413 - 0.5000 = 0.3413$
c. $P(X \geq 7) = P\left(\frac{X-3}{2} \geq \frac{7-3}{2}\right) = P(Z \geq 2) = 1 - P(Z < 2) = 1 - 0.9772 = 0.0228$

Exercise 5.3 (Hemoglobin)**.** The hemoglobin (Hb) value in a male population is normally distributed with mean 188 g/L and standard deviation 14 g/L.

a) Men with Hb below 158 g/L are considered anemic. What is the probability of a random man being anemic?
b) When randomly selecting 10 men from the population, what is the probability that none of them are anemic?

Solution

a) $P(Hb < 158) = P(Z < \frac{158-188}{14}) = P(Z < -2.14) = [table] = 0.016$ or use R

```
pnorm(158, mean=188, sd=14)
```

```
[1] 0.01606229
```

b) Probability of one man not being anemic; $1 - 0.016 = 0.984$. Probability of 10 selected men not being anemic (binomial distribution) $0.984^{10} = 0.85$

Exercise 5.4 (Pill)**.** A drug company is producing a pill, with on average 12 mg of active substance. The amount of active substance is normally distributed with mean 12 mg and standard deviation 0.5 mg, if the production is without problems. Sometimes there is a problem with the production and the amount of active substance will be too high or too low, in which case the pill has to be discarded. What should the upper and lower critical values (limits for when a pill is acceptable) be in order not to discard more than 1/20 pills from a problem free production?

Solution

$H_0: \mu = 12$ $H_1: \mu \neq 12$
Search x_{low} and x_{up} such that $P(x_{low} < X < x_{up}) = 0.95$
From table we know that $P(-1.96 < Z < 1.96) = 0.95$

$Z = \frac{X-\mu_0}{\sigma_0} = \frac{X-12}{0.5}$, hence
$-1.96 < Z < 1.96 \iff -1.96 < \frac{X-12}{0.5} < 1.96 \iff 12 - 1.96 * 0.5 < X < 12 + 1.96 * 0.5 \iff 11.02 < X < 12.98$
The lower and upper critical values of active substance should be 11.02 and 12.98 mg.

Random sample

Exercise 5.5 (Exercise in distribution of sample mean)**.** The total cholesterol in population (mg/dL) is normally distributed with $\mu = 202$ and $\sigma = 40$.

a. How is the sample mean of a sample of 4 persons distributed?
b. What is the probability to see a sample mean of 260 mg/dL or higher?
c. Is there reason to believe that the four persons with mean 260 mg/dL were sampled from another population with higher population mean?

Solution

a.

$$\bar{X} = \frac{1}{4}\sum_{i=1}^{4} X_i X_i \sim N(\mu, \sigma^2)\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N(202, 1600/4) = N(202, 400)$$

b. 0.0019
c. Yes, sample mean of 260 is quite unlikely under the stated population model, so this would provide evidence that the sample may come from a population with a higher mean.

Exercise 5.6 (Amount of active substance)**.** The amount of active substance in a pill is stated by the manufacturer to be normally distributed with mean 12 mg and standard deviation 0.5 mg. You take a sample of five pills and measure the amount of active substance to; 13.0, 12.3, 12.6, 12.5, 12.7 mg.

[Note: a-c were already computed in the descriptive statistics session.]

a) Compute the sample mean
b) Compute the sample variance
c) Compute the sample standard deviation
d) compute the standard error of mean, SEM.
e) If the manufacturers claim is correct, what is the probability to see a sample mean as high as in (a) or higher?

Answer

a. 12.62
b. 0.067
c. 0.26
d. 0.22
e. 0.0028

Solution

```
x <- c(13.0, 12.3, 12.6, 12.5, 12.7)
n <- length(x)
(m <- sum(x)/n)
```

```
[1] 12.62
```

b. Sample variance

```
[1] 0.067
```

c. Sample standard deviation

```
[1] 0.2588436
```

d. Standard error of mean, SEM

```
[1] 0.2236068
```

Note, here the known standard deviation, $\sigma = 0.5$ is used.

e. Probability to see a sample mean as high as in (a) or higher

```
[1] 0.00277946
```

References